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Postdoctoral Researcher

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I am a Postdoctoral Researcher at the **Georgia Institute of Technology**, specializing in **batteryless sensing platforms, energy harvesting, sustainable embedded systems, and energy-efficient embedded AI**. My research advances the foundations of ultra-low-power and intermittent computing through resilient hardware/software co-designs, adaptive energy-aware runtimes, and efficient machine learning techniques for resource- and energy-constrained devices. By co-designing computing, energy management, and intelligence around highly variable energy availability, I develop embedded systems that can operate reliably and intelligently under real-world constraints.

I have authored 10+ peer-reviewed publications in top-tier venues such as **OSDI, EuroSys, SenSys, EWSN, and MobiSys**. I was selected for the **EWSN 2025 Rising Stars Forum** and the **MobiSys 2026 Rising Stars Forum**. My work spans sustainable sensing and energy-efficient embedded intelligence for applications including smart infrastructure, healthcare, and environmental monitoring, with the broader goal of enabling intelligent edge systems that operate within stringent energy and sustainability constraints.

EXPERTISE

Embedded Systems

Operating Systems/Runtimes

Computer Architectures

Low-power Architectures

Energy Harvesting

Intermittent Computing

Tiny Machine Learning

Sustainable IoT Systems

PUBLICATION RECORD

Selected Publications

■ **SenSys 2027**: *"Failure Is an Option": Broken Memory as a Timekeeper for Intermittent Computing*. Nikola Maruszewski, Jordan Athas, Allison Fleming, Christian Duffee, **Eren Yildiz** et al. ACM Conference on Embedded Networked Sensor Systems (Accepted).

■ **MobiSys 2026**: *BIONIC: Power System Architecture for Time-Sensitive Intermittent Computing*. **Eren Yildiz** et al. The ACM International Conference on Mobile Systems, Applications, and Services.

■ **MobiSys 2026**: *MicroGreen: A Framework on Carbon-aware Edge ML System Design*. Xuesi Chen, Ilan Mandel, **Eren Yildiz** et al. The ACM International Conference on Mobile Systems, Applications, and Services.

■ **EWSN 2025**: *PEARL: Power- and Energy-Aware Multicore Intermittent Computing*. Khakim Akhunov, **Eren Yildiz** et al. The 22nd International Conference on Embedded Wireless Systems and Networks.

■ **SenSys 2024**: *Fast-Inf: ultra-fast embedded intelligence on the batteryless edge*. Leonardo Lucio Custode, Pietro Farina, **Eren Yildiz** et al. The 22nd ACM Conference on Embedded Networked Sensor Systems.

■ **EuroSys 2024**: *Adaptable runtime monitoring for intermittent systems*. **Eren Yildiz** et al. The 19th European Conference on Computer Systems.

■ **EWSN 2024**: *Memory-efficient energy-adaptive inference of pre-trained models on batteryless embedded systems*. Pietro Farina, Subrata Biswas, **Eren Yildiz** et al. The 21st International Conference on Embedded Wireless Systems and Networks.

■ **EuroSys 2023**: *Efficient and safe i/o operations for intermittent systems*. **Eren Yildiz** et al. The 18th European Conference on Computer Systems.

■ **OSDI 2022**: *Immortal threads: Multithreaded event-driven intermittent computing on Ultra-Low-Power microcontrollers*. **Eren Yildiz** et al. In 16th USENIX Symposium on Operating Systems Design and Implementation.

Current Research Pipeline

■ **Patent**: *A New Power System Architecture for Batteryless Systems*. **Eren Yildiz** et al. Italian Patent and Trademark Office (Under Evaluation).

📄 **Patent:** A spintronic timekeeper circuit and system for ultra-low power embedded applications. *Eren Yildiz* et al. United States Patent and Trademark Office (**Under Evaluation**).

📖 **SenSys 2027:** Event-Driven Programming Models for Intermittent Systems. *Eren Yildiz* et al. ACM Conference on Embedded Networked Sensor Systems (**Target: SenSys 2027 Fall Submission**).

📖 **SenSys 2027:** Adaptive Intelligence for Energy-Harvesting Edge Systems. *Eren Yildiz* et al. ACM Conference on Embedded Networked Sensor Systems (**Target: SenSys 2027 Fall Submission**).

📖 **SenSys 2027:** Efficient Model Compression for Embedded AI. *Eren Yildiz* et al. ACM Conference on Embedded Networked Sensor Systems (**Target: SenSys 2027 Fall Submission**).

RESEARCH ARTIFACTS

- 🔗 **Immortal Threads:** <https://github.com/tinysystems/ImmortalThreads>
Immortal Threads is a novel programming model that brings pseudo-stackful multithreaded processing to intermittent computing. Programmers using Immortal Threads are oblivious to intermittent execution and write their applications in a multithreaded fashion using common event-driven multithreading primitives.
- 🔗 **EaselIO:** <https://github.com/tinysystems/easelIO>
EaselIO is a novel programming model that introduces re-execution semantics of IO operations for intermittent computing. Re-executed IO operations in intermittent systems' tasks might lead to memory inconsistencies and high energy consumption. EaselIO offers new interfaces to call IO functions to overcome memory inconsistencies and save energy.
- 🔗 **ARTEMIS:** <https://github.com/tinysystems/ARTEMIS>
ARTEMIS is the first framework designed to facilitate flexible property checking of intermittent programs at runtime. In ARTEMIS, developers use a language to define desired properties of the intermittent system, focusing on system properties rather than implementation specifics.
- 🔗 **PEARL:** <https://github.com/tinysystems/PEARL>
PEARL is a novel systems support that can make existing multicore microcontroller (MCU) platforms suitable for efficient intermittent computing.
- 🔗 **FreeML:** <https://github.com/tinysystems/FreeML>
FreeML is a framework to optimize pre-trained DNN models for memory-efficient and energy-adaptive inference on batteryless systems.

SERVICE RECORD

📁 Conference/Workshop TPCs

- ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (**SenSys'27**)
- 14th International Workshop on Energy Harvesting & Energy-Neutral Sensing Systems (**ENSys'26**)

📁 Journal Reviews

- ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies
- IEEE Internet of Things Journal
- Future Generation Computer Systems
- IEEE Transactions on Very Large Scale Integration Systems
- IEEE Transactions on Sustainable Computing
- IEEE Transactions on Mobile Computing
- IEEE Open Journal of the Communications Society

📁 As an External Reviewer

- The ACM SIGBED International Conference on Embedded Software (**EMSOFT**)
- The ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies (**COMPASS 2026**)

📁 Other

- Session Chair, Rising Stars Forum, The ACM International Conference on Mobile Systems, Applications, and Services (**MobiSys'26**)

ACHIEVEMENTS & AWARDS



MobiSys Rising Star 2026.

I selected as one of the Rising Star in the 24th ACM International Conference on Mobile Systems, Applications, and Services (MobiSys) 2026.



EWSN Rising Star 2025.

I selected as one of the Rising Star in Conference on Embedded Wireless Systems and Networks (EWSN) 2025.



Research Fellowship Award

2214-A - International Research Fellowship Programme for PhD Students. The Scientific And Technological Research Council Of Turkey(TUBITAK), 20,000\$.

EMPLOYMENT AND APPOINTMENTS



Postdoctoral Researcher

📅 Feb. 2025 – Ongoing

Georgia Institute of Technology, College of Computing, School of Interactive Computing, Atlanta, United States



Research Assistant Dr.

📅 Oct. 2023 – Feb. 2025

RTEU, Department of Computer Engineering, Rize, Turkiye



Visiting Researcher

📅 Jul. 2024 – Aug. 2024

University of Trento, Department of Information Engineering and Computer Science, Trento, Italy



Visiting Researcher

📅 Dec. 2021 – Dec. 2022

Northwestern University, Computer Science Department, Chicago, United States



Research Assistant

📅 Oct. 2018 – Oct. 2023

Ege University, Department of Computer Engineering, Izmir, Turkiye



R&D Engineer

📅 Sept. 2015 – Feb. 2017

Emtech Information Technologies, Mersin, Turkiye

EDUCATION

Ph.D. in Computer Engineering

Ege University

📅 2018 – 2023

Thesis title: System Support for Efficient Operation of Batteryless Sensor Systems

M.Sc. in Electrical Electronics Engineering

Mersin University

📅 2015 – 2018

Thesis title: Cyberbullying Detection And Classification With Artificial Intelligence Algorithms In Social Network

B.Sc. in Electrical Electronics Engineering

Mersin University

📅 2009 – 2014

COURSES TAUGHT

Courses at the Department of Computer Engineering, RTEU, Turkiye



CE-202 Signals and Systems

📅 2024 - Spring Semester



CE-414 Tiny Machine Learning

📅 2024 - Spring Semester



CEN-109 Fundamental Information Technologies

2024 - Fall Semester

TEACHING ASSISTANT EXPERIENCE

Courses at the Department of Computer Engineering, Ege University, Turkiye



Digital Computer Design

2019 - 2023 Spring Semester



Logic Design

2019 - 2023 Fall Semester



Digital Electronics

2019 - 2023 Spring Semester



Electrical Circuits

2019 - 2023 Fall Semester



Microprocessors

2019 - 2023 Fall Semester



Microcontroller-Based System Design-

2019 - 2023 Spring Semester

MENTORING

PhD Students Mentoring

- **Summit Shrestha (Georgia Tech):** Mentoring research on energy- and carbon-efficient LLM execution on edge and intermittent devices.
- **Nikola Maruszewski (Georgia Tech):** Mentoring research on MTJ-based spintronic elements for sustainable, ultra-low-power IoT architectures.

Master Students Mentoring

- **Pichaya Limprayoon (Georgia Tech):** Mentoring research on low-power timekeeping mechanisms for battery-free and intermittent systems.
- **Amritha Pramod (Georgia Tech):** Mentoring research on robust and resource-efficient embedded AI under intermittent and incomplete input data.

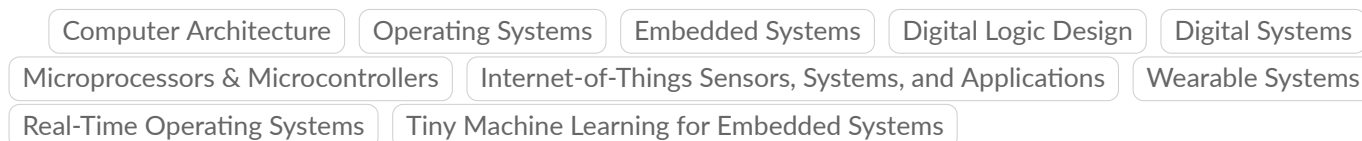
Undergraduate Students Mentoring

I have supervised several undergraduate students whose work has resulted in high-impact outcomes.

- **Davide Cavedon & Stefano Putelli (University of Trento):** Co-developed a new energy storage and management architecture for batteryless systems, leading to an **MobiSys'26 submission and a patent submission**.
- **Pietro Farina (University of Trento):** Worked on TinyML optimization for intermittent systems, **resulting in a publication at EWSN'24**.
- **Lijun Chen (University of Trento):** Worked on compiler front-end design of Immortal Threads runtime, **resulting in a publication at OSDI'22**.

This demonstrates my ability to guide students at different levels toward meaningful contributions, including top-tier submissions, patents, and interdisciplinary research directions.

TEACHING INTERESTS



PROFESSIONAL MEMBERSHIPS



HiPEAC (High Performance Embedded Architecture and Compilation)

 <https://www.hipeac.net>

Publications

Journal Articles

- X. Chen, A. Goldner, E. Yildiz, *et al.*, “From component to system: Rethinking edge computing design through a carbon-aware lens,” [ACM SIGENERGY Energy Informatics Review](#), vol. 5, no. 2, pp. 125–131, 2025.
- M. Doglioni, E. Yildiz, M. Nardello, K. Akhunov, K. S. Yildirim, and D. Brunelli, “Capdyn: Adaptive self-scaling energy storage for powering batteryless iot,” [ACM Transactions on Embedded Computing Systems \(TECS\)](#), 2025.
- F. Erata, E. Yildiz, A. Goknil, *et al.*, “Etap: Energy-aware timing analysis of intermittent programs,” [ACM Transactions on Embedded Computing Systems \(TECS\)](#), 2022.
- A. Cigdem, E. Yildiz, and E. S. Essiz, “Automatic detection of cyberbullying in formspring. me, myspace and youtube social networks,” [Turkish Journal of Engineering](#), vol. 3, no. 4, pp. 168–178, 2019.

Conference Proceedings

- X. Chen, I. Mandel, E. Yildiz, J. Hester, and U. Gupta, “A greener edge: A framework on carbon-aware edge ml system design,” in [ACM International Conference on Mobile Systems, Applications, and Services](#), **MobiSys’26**, 2026.
- E. Yildiz, D. Cavedon, S. A. Putelli, J. Hester, and K. S. Yildirim, “Bionic: A co-designed hardware and runtime for time-sensitive battery-free iot,” in [ACM International Conference on Mobile Systems, Applications, and Services](#), **MobiSys’26**, 2026.
- K. Akhunov, E. Yildiz, and K. S. Yildirim, “Pearl: Power- and energy-aware multicore intermittent computing,” in [Proceedings of the 2025 International Conference on Embedded Wireless Systems and Networks](#), **EWSN’25**, 2025.
- S. Ahmed, E. Yildiz, S. Holla, *et al.*, “Bootstrapping health wearables powered by intra-body power transfer,” in [2024 IEEE 20th International Conference on Body Sensor Networks \(BSN\)](#), IEEE, 2024, pp. 1–4.
- L. L. Custode, P. Farina, E. Yildiz, R. B. Kilic, K. S. Yildirim, and G. Iacca, “Fast-inf: Ultra-fast embedded intelligence on the batteryless edge,” in [Proceedings of the 22nd ACM Conference on Embedded Networked Sensor Systems](#), **SensSys’24**, 2024, pp. 239–252.
- P. Farina, S. Biswas, E. Yildiz, *et al.*, “Memory-efficient energy-adaptive inference of pre-trained models on battery-less embedded systems,” in [Proceedings of the 2024 International Conference on Embedded Wireless Systems and Networks \(EWSN’24\)](#), **EWSN’24**, 2024.
- E. Yildiz, K. Akhunov, L. A. Riva, A. Goknil, I. Kurtev, and K. S. Yildirim, “Adaptable runtime monitoring for intermittent systems,” in [Proceedings of the Nineteenth European Conference on Computer Systems \(EuroSys’24\)](#), **EuroSys 24**, 2024, pp. 1175–1191.
- E. Yildiz, D. Cavedon, and K. S. Yildirim, “On tracking time with better resolution and range in batteryless systems,” in [Proceedings of the 12th International Workshop on Energy Harvesting and Energy-Neutral Sensing Systems \(EnsSys’24\)](#), 2024, pp. 42–47.
- K. Akhunov, E. Yildiz, and K. S. Yildirim, “Enabling efficient intermittent computing on brand new microcontrollers via tracking programmable voltage thresholds,” in [Proceedings of the 11th International Workshop on Energy Harvesting & Energy-Neutral Sensing Systems \(EnsSys’21\)](#), 2023, pp. 16–22.
- E. Yildiz, S. Ahmed, B. Islam, J. Hester, and K. S. Yildirim, “Efficient and safe i/o operations for intermittent systems,” in [Proceedings of the Eighteenth European Conference on Computer Systems](#), **EuroSys 23**, 2023.

- E. Yildiz, L. Chen, and K. S. Yildirim, "Immortal threads: Multithreaded event-driven intermittent computing on {ultra-low-power} microcontrollers," in [16th USENIX Symposium on Operating Systems Design and Implementation, OSDI 22](#), 2022, pp. 339–355.
- E. Yildiz and K. S. Yildirim, "Persistent timekeeping using harvested power measurements," in [Proceedings of the 9th International Workshop on Energy Harvesting and Energy-Neutral Sensing Systems\(EnsSys'21\)](#), 2021, pp. 572–574.
- E. Yildiz and K. S. Yildirim, "Defragmenting energy storage in batteryless sensing devices," in [Proceedings of the 8th International Workshop on Energy Harvesting and Energy-Neutral Sensing Systems\(EnsSys'20\)](#), 2020, pp. 36–42.
- E. Yildiz, K. S. Yildirim, P. Pawelczak, and J. Hester, "On the accuracy of network synchronization using persistent hourglass clocks," in [Proceedings of the 7th International Workshop on Energy Harvesting & Energy-Neutral Sensing Systems \(EnsSys'19\)](#), 2019, pp. 35–41.
- E. Yildiz, C. Aci, and E. S. Essiz, "The effects of attribute selection in artificial neural network based classifiers on cyberbullying detection," in [2018 3rd International Conference on Computer Science and Engineering \(UBMK\)](#), IEEE, 2018, pp. 6–11.
- E. Yildiz, Ç. Aci, and E. S. Eşsiz, "Performance analysis of artificial neural network based classifiers for cyberbullying detection," in [2018 3rd International Conference on Computer Science and Engineering \(UBMK\)](#), IEEE, 2018, pp. 1–5.

References

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